

AMD's Athlon 64 X2 4800+ and Intel's Pentium D 820

Duel Of The Dual-Cores!

We've been talking about dual-core processors for some time now, giving truckloads of *gyan* about how two cores are better than one and how this will usher in a new era in desktop computing. We managed to grab hold of two of these—one each from AMD and Intel—and we put them to the test to see if all the hype is really worth anything.

We would like to make it clear at the outset that although this is not a head-to-head comparison, the Athlon 64 X2 is the best that AMD has to offer. And the Pentium D 820 is the lowest dual-core processor from Intel, and is aimed at the budget-conscious consumer.

The tests we conducted are an indication of the kind of performance that can be expected from these processors. First of all, though, we will shed a little light on what lies inside each of the CPUs.

The Athlon 64 X2 4800+ is a 939-pin processor and is fabricated using the 90 nm technology. It is clocked at 2.4 GHz and has 1024 KB of L2 cache. It fits on any existing Socket 939 motherboard, though a BIOS update might sometimes be necessary.

The X2 4800+ is essentially two K8 cores of Athlon 64 4000+ processors on a single chip and hence, a model number increment of just 800 might seem a bit too conservative.

It has all the popular instruction sets such as MMX, 3DNow!, SSE, SSE2 and SSE3 (it has all SSE3 instructions except those pertaining to HyperThreading, an Intel-specific feature). The SSE3 has been incorporated into the Athlon 64 line since the revision E3 or the Venice core.

Intel's Pentium D 820 is an LGA 775 processor and is also an 90 nm part. It runs at 2.8 GHz and has a L2 cache of 1 MB. It fits on almost any LGA 775 motherboard, but this particular model has some issues with the nForce4 chipset. The Pentium D 820, based on the Smithfield core, is in fact two Prescott cores on a single chip.

These are the same as two Pentium IV Extreme Edition CPUs stuck together. The only difference is that these cores are not HyperThreaded, which means that each core runs a single thread.

Unlike the 820, the Pentium Extreme Edition 840 has two HyperThreaded cores which translate to four threads running at the same time. It is needless to add that both processors have 64-bit architecture and native support for 64-bit operating systems as well as applications.

To run the benchmarks, the Athlon 64 X2 was mounted on a motherboard with NVIDIA nForce4 Ultra chipset with 1 GB DDR RAM whereas the Pentium D 820 was mounted on an Intel 955X based motherboard with 1 GB DDR2 RAM. All other components were the same.

In the gaming tests which included *Doom 3* and *FarCry*, the X2, along with the GeForce GT, scored fluid frame rates while in 3DMark05, the X2 scored 6,792 CPU marks. This means that the X2 will be definitely get you closer to gaming nirvana!

In the CineBench 2003 test, where we evaluate multi-threaded 3D rendering, the result was no different with the X2. This proved that the X2 has raw strength to efficiently handle 3D applications such as AutoCAD, Maya and so on.

In the number crunching benchmarks from ScienceMark, SiSoft Sandra and Everest, the X2 proved its mettle by racing ahead of every processor we had tested so far, proving its supremacy in computational power. Finally, in the Multimedia Content Creation benchmark, the X2 scored a huge 38.8, the highest so far, which shows that it is well suited to handling multimedia applications like 3D Studio Max and Adobe Premiere to name a couple.

On the Intel front, the Pentium D 820 was no wimp, and it scored frame rates good enough for most

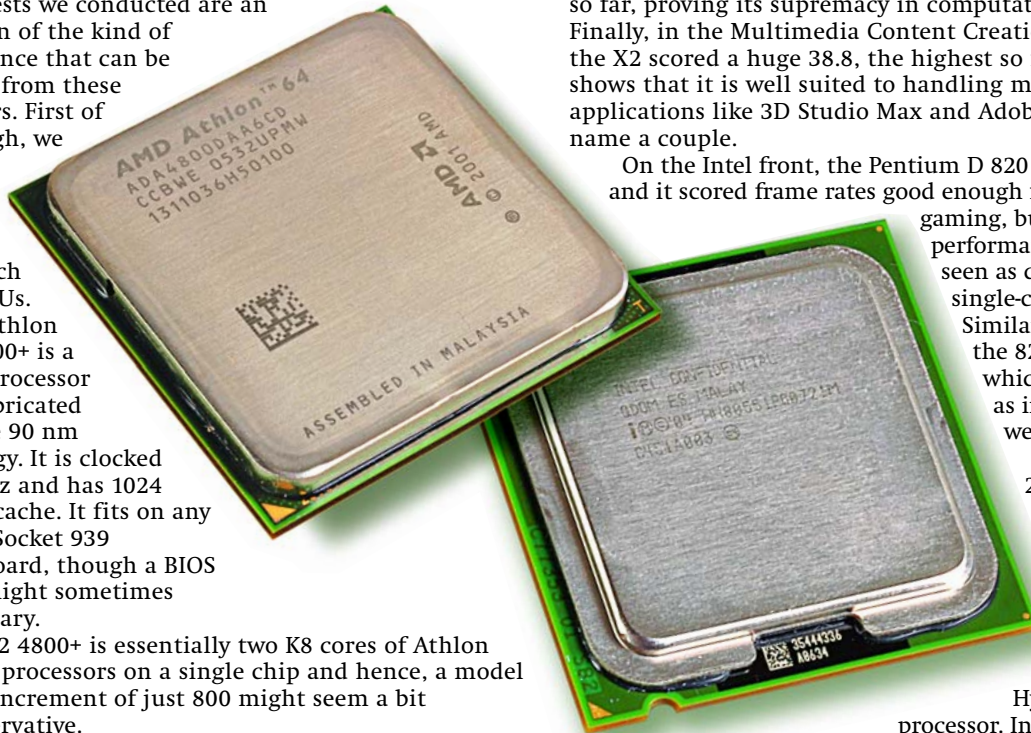
gaming, but not much performance gain was seen as compared to its single-core counterpart. Similarly in 3DMark, the 820 scored 5,385, which again, was not as impressive as we'd hoped.

In CineBench 2003, the 820 once again performed slightly better than a normal

HyperThreaded processor. In the number crunching benchmarks, though, it was ahead of the single-core Pentium IV by a comfortable margin. In the Multimedia Content Creation test, it scored 27.9, which is just a little better than the P IV.

The Athlon 64 X2 4800+ is the highest offering from AMD, and is in direct competition with the Intel Pentium Extreme Edition 840. On the other hand, the Pentium D 820 is in a strange three-way-fight against the lower priced Athlon 64 X2 4200+ as well as the single core Athlon 64 4000+, and it is a little unfair to compare it with the monster that the Athlon 64 X2 4800+ is.

Currently, only the price is in the favour of the Intel Pentium D 820: its AMD counterpart is much more expensive. If AMD doesn't lower the prices of its low-end dual-core processors, budget-conscious enthusiasts will be forced to buy the mediocre but affordable Intel Pentium D 820.

**AMD Athlon 64x2 4800**

Company **AMD Far East Ltd (India)**
Phone **080-51372400**
Web site **www.amd.com**

Intel Pentium D820

Company **Intel Technology India Pvt Ltd**
Phone **080-25075000**
Web site **www.intel.com**